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## **Synergistic Oil Recovery Effect of Nanoparticles and Surfactants**

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### **Abstract**

Due to being environmentally friendly, nanoparticles have recently attracted lots of attention for enhanced oil recovery applications. However, the main challenge in using nanoparticles is that they must maintain colloidal stability under harsh reservoir conditions. This research focuses on assessing surface-modified fumed and colloidal nano silicas on surfactants to enhance oil recovery, with a special focus on wettability alteration, colloidal stability in brine at high temperature, interfacial tension (IFT), and emulsification characteristics.

Contact angle measurements were used to study the wettability alteration of core chips using the captive bubble method. Interfacial tension was measured by using the spinning drop method. Recovering of liquid hydrocarbons was tested by spontaneous imbibition experiments. The emulsion tendency was studied visualizing the phase behavior of the oil and aqueous solution after shaking the sample for 30 seconds. The surface of the silica nanoparticles was modified with silane to enhance the colloidal stability of nanofluids.

Surface-modified nano silicas did not change the core wettability and remained oil wet while the surfactant altered to intermediate wet. However, the mixture of nano silica and surfactant changed the core wettability to water wet, exhibited a lower IFT than individual components, and produced 60% oil recovery as compared to 20% for surfactant alone and 11% for particles alone, suggesting the significant synergistic mixture. Colloidal stability was achieved and then enhanced when blended with a small amount of alum. Emulsions were resolved after 20 minutes when treated with the mixture of nano silica and surfactant compared to stable emulsion for the untreated.

Surface-modified nano silicas, in general, can deliver significant benefits, including reduced chemical usage, optimized logistics, and lower CO<sub>2</sub> emissions.

### **Introduction**

Nanotechnology is an emerging technology that has the potential to change the technology landscape of the oil and gas industry (Ko and Huh, 2019; Fakoya and Shah, 2017), especially in enhanced oil recovery (EOR) (Zhao et al., 2019). Nanoparticles (NP) have unique properties due to their small size and high surface area per unit volume, thus resulting in higher reactivity with other molecules such as surfactants. NP can alter the reservoir formations from being oil-wet to water-wet, and decrease the disjoining pressure, leading to increased oil production. A variety of NP based on silica (Alzobaidi et al., 2021), halloysite (Zhao et al.,

2019),  $\text{Al}_2\text{O}_3$  (Nowrouzi et al., 2022), graphene oxide (Luo et al., 2016), and  $\text{TiO}_2$  (Nowrouzi et al., 2022) have recently been found to be quite effective at wettability alteration and improving oil recovery. However, the brines in these studies typically have low total dissolved solids (TDS). Tang et al., (2023) observed a synergy of surface modified nanoparticles and cationic surfactants in wettability of calcite.

Synergistic mixtures of nanoparticles and surfactants are also a recent topic of interest. Zao et al. (2023) reported an oil recovery of 30% using a synergistic blend of sulfonated silicon quantum dots NP and betaine surfactant when compared to 13% and 17% for the NP alone and surfactant alone, respectively. Al-Shatty et al. (2022) showed that a synergistic blend of hydrophobic octanoic acid-coated alumina particles and cationic surfactant enhances the oil recovery 5% compared to surfactant alone. Tarek M and El-Banbi A.H (2015) used mixtures of silica, aluminum oxide, and iron oxide suspended in water to enhance oil recovery, and they offer benefits over single nano-fluids. Silica alters the rock wettability. Aluminum oxide lowers the oil viscosity by breaking the carbon-sulfur bonds. Iron oxide increases the density and viscosity of the injected water and as a result the sweep efficiency is increased. Pei et al. (2015) observed a mixture of silica and cationic surfactant emulsified heavy oil in water to form emulsions, leading to enhanced oil recovery. High adsorption onto rock surfaces is one of the major issues with surfactants, rendering them expensive and less efficient. NP can reduce the adsorption of surfactants on the rock surface which increases the effectiveness in terms of oil production. Al-Anssari et al. (2017) reported mixtures of sodium dodecyl sulfate (SDS) and silica nanoparticles altering the limestone rock to strongly water-wet and increased the oil recovery when compared to individual component. This synergistic effect can be attributed to adsorption of nano silica particles on carbonate surfaces, reducing the loss of anionic surfactant molecules in the porous media and rendering enough anionic surfactant molecules available to contact the oil. Zargartalebi et al. (2015) found that silica nanoparticle reduces the adsorption of an anionic surfactant on sandstone rock surface. By using  $\text{TiO}_2$ , the adsorption of a mixture of surfactants on crushed sandstone grains can be reduced by half of the adsorption value of the surfactants alone, causing an increase in the oil recovery (Nourafkan et al., 2018). Oppositely charged surfactants may adsorb onto nanoparticles, causing them hydrophobic and facilitating their adsorption at the oil-water interface. This would lead to emulsification, resulting in enhancing oil recovery (Zaid et al., 2012). Or like-charged nanoparticles can further increase surfactant effectiveness at oil-water interface and rock-water interface, leading to increased oil recovery (Vatanparast et al., 2018).

The observed problems associated with the use of NP in the literature are their instability in harsh reservoir conditions (high salinity and high temperature conditions) and their retention on reservoir rocks, causing a decrease in permeability by clogging pore throats. In this study, a surfactant was used with surface-modified NP. The objectives of this study were to investigate: (1) the colloidal stability of the surface modified silica NP under harsh reservoir conditions with high salinity in the presence of divalent ions and high temperature; (2) the wettability alteration; (3) interfacial tension reduction and (3) the synergy between the NP and surfactants in increasing oil production under the reservoir conditions.

## Experimental Materials and Methods

### Materials

ChampionX commercial surfactant (S) and a new functionalized fumed nano silica (FNS) were used for the study. FNS has a median particle size of 114 nm, a surface area of 170  $\text{m}^2/\text{g}$ , and a zeta potential of -15 mV at pH 7. Functionalized colloidal silica nanoparticles (FCS) were made by ChampionX from native colloidal nano silicas (NCS) by reacting with a nonionic silane coupling agent at different ratios. Tables 1 and 2 list the properties of NCS and FCS.

**Table 1—Native colloidal silica (NCS) properties**

| Sample ID | %actives | Surface area(m <sup>2</sup> /g) | pH    | Particle size (nm) |
|-----------|----------|---------------------------------|-------|--------------------|
| NCS1      | 30.11    | 331.10                          | 10.06 | 9.06               |

**Table 2—Functionalized colloidal silica (FCS) properties**

| Sample ID | Silica type | %actives | pH     | μmoles siloxane/m <sup>2</sup> | Surface area(m <sup>2</sup> /g) |
|-----------|-------------|----------|--------|--------------------------------|---------------------------------|
| FCS6      | NCS1        | 11.38    | 10.006 | 10                             | 153.53                          |
| FCS7      | NCS1        | 11.72    | 10,07  | 15                             | 113.48                          |

The oil used in all tests was from an unconventional field in the Permian Basin which has an API gravity of 42 at 15°C and Table 3 shows the properties (Table 3):

**Table 3—Oil Properties**

| Sample ID                | Oil Composition (wt%) |       |      |      |
|--------------------------|-----------------------|-------|------|------|
|                          | SAT                   | ARO   | NSO  | ASPH |
| Unconventional Field Oil | 80.40                 | 17.63 | 1.88 | 0.09 |

Table 4 shows the recipe of a synthetic brine.

**Table 4—Synthetic Brine Recipe**

| Salt                                              | gm/Lit  |
|---------------------------------------------------|---------|
| CaCl <sub>2</sub> (H <sub>2</sub> O) <sub>2</sub> | 4.035   |
| MgCl <sub>2</sub> (H <sub>2</sub> O) <sub>6</sub> | 1.338   |
| NaCl                                              | 117.348 |
| KCl                                               | 1.316   |
| NaBr                                              | 0.773   |
| SrCl <sub>2</sub>                                 | 0.822   |

Core plugs were tested from an unconventional field in the Permian Basin.

## Methods

1. Stability at reservoir temperature in synthetic brine for 30 days: Samples are made at various concentrations in the brine of interest and placed into an oven at reservoir temperature. Samples are observed over the course of 30 days for signs of instability. A positive result would be a clear, homogenous solution with no signs of particulate or solids settling out or layers.
2. Contact angle/wettability alteration measurements: Chips from core samples are polished using a diamond plate to ensure a smooth surface. Smoothed chips are vacuum saturated with oil then aged at reservoir temperature as per the standard operating procedure. The saturated and aged chips are then placed into a solution of brine or brine plus Production Enhancement (PE) solution or PE plus nanoparticles at the desired dose rate and aged at reservoir temperature for approximately three days. The chip is mounted in a cell, and the solution of interest is added to the cell containing the chip. A drop of oil is placed on the underside of the chip and the contact angle is then measured by the goniometer software.

3. Interfacial Tension measurements: Interfacial tension is measured using a Spinning Drop Tensiometer (SDT). Interfacial tension is measured by analyzing a video image of a drop of oil surrounded by the fluid of interest in a rotating capillary. IFT is measured at ambient laboratory temperature as well as reservoir temperature if possible.
4. Spontaneous Imbibition determination: A saturated and aged core is placed in an imbibition cell such that none of the surfaces are in contact with the cell walls or floors. The test solution of interest (for example, brine, PE solution or PE plus nanoparticles) is added to the cell such that the liquid advances up into the neck of the cell with the volume markings. The cell is then placed into an oil bath or oven at bottom hole temperature and the amount of oil that floats to the top of the cell is measured and plotted. The test is run until oil is no longer produced.
5. Emulsion tendency determination: Equal parts of crude oil and either brine, PE or PE plus nanoparticles at the desired dose rate are placed into a tube. The tubes are heated to reservoir temperature. The tubes are mixed and then allowed to sit undisturbed and are observed for phase separation.

## Results and Discussion

### Colloidal stability in Permian brine

The sum of attractive and repulsive forces between nanoparticles dictates aggregation of nanoparticles. When the attractive force is less than the repulsive force, particle aggregation can be prevented, enhancing colloidal stability. It has been reported that the addition of a silane coupling agent to modify the surface of nano silicas improves the colloidal stability of nanofluids ([Mukherjee and Paria, 2013](#)). In this study, the surface of the silica nanoparticles is modified with a nonionic silane to increase the repulsive force via steric stabilization between particles. The ratio of silane to silica played a significant role in colloidal stability.

**Functionalized fumed silica (FSF).** Initial testing was done at 1000 ppm total dosage based on active material, which is the typical dosage for PE products. Various ratios of surfactant (S) to FSF were tested and observed over the course of 31 days at 70°C. The results can be seen in [Figure 1](#) below. Over the course of 31 days at 70°C, the mixtures were stable in brine.

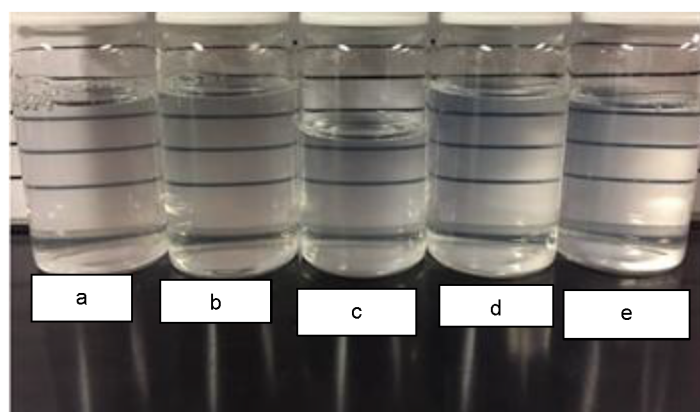


Figure 1—1000 ppm total dose of various ratios of S to FSF in synthetic brine: (a) 9:1; (b) 3:1; (c) 1:1; (d) 1:3; (e) 1:9

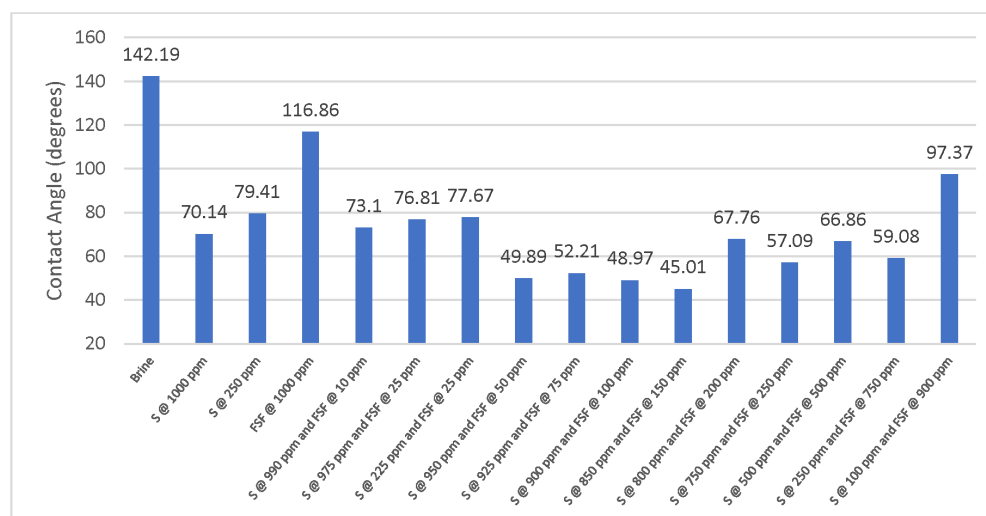
**Functionalized colloidal silica (FCS).** Stability tests were done on various ratios of the blended and non-blended siloxane functionalized nanosilica particles. Testing was done at 75°C for 30 days using synthetic brine as the dilution water. The results can be seen in [Table 5](#) below. The stability tests show that FS 7 blended with Alum is the most stable of the various products tested. The blended versions of FCS 6 and FCS 7 with Alum will have a B at the end of the sample name.

**Table 5—Stability test results of various blends of S and functionalized colloidal nanosilica particles (FCS) at 75°C**

| Treatment                                   | Result             |
|---------------------------------------------|--------------------|
| 1000 ppm FCS 6                              | Unstable – 7 days  |
| 1000 ppm FCS 6 Blend w/Alum                 | Stable – 30 days   |
| 1000 ppm FCS 7                              | Stable – 30 days   |
| 1000 FCS 7 Blend w/Alum                     | Stable – 30 days   |
| 900 ppm S and 100 ppm FCS 6                 | Unstable – 7 days  |
| 900 ppm S and 100 ppm FCS 6 blend with Alum | Unstable – 10 days |
| 900 ppm S and 100 ppm FCS 7                 | Unstable – 10 days |
| 900 ppm S and 100 ppm FCS 7 Blend w/Alum    | Unstable – 10 days |
| 450 ppm S and 50 ppm FCS 6                  | Unstable – 10 days |
| 450 ppm S and 50 ppm FCS 6 Blend w/Alum     | Stable – 30 days   |
| 450 ppm S and 50 ppm FCS 7                  | Stable – 30 days   |
| 450 ppm S and 50 ppm FCS 7 Blend w/Alum     | Stable – 30 days   |
| 225 ppm S and 25 ppm FCS 6                  | Unstable – 10 days |
| 225 ppm S and 25 ppm FCS 6 Blend w/Alum     | Stable – 30 days   |
| 225 ppm S and 25 ppm FCS 7                  | Stable – 30 days   |
| 225 ppm S and 25 ppm FCS 7 Blend w/Alum     | Stable – 30 days   |

## Contact Angle/Wettability Alteration Test Results

**Functionalized fumed silica (FSF).** The contact angle test results for various ratios of a surfactant (S) and FSF in brine can be seen in Figure 2.

**Figure 2—Contact Angle Results for Various Ratios of S and FSF in synthetic brine**

The following criteria are used when determining the state of the reservoir with regards to contact angle:  $0 < \theta < 70$  is water wet,  $70 < \theta < 110$  is intermediate and  $110 < \theta < 180$  is an oil wet system. The results from the contact angle test show the following: 1) both brine and FSF exhibit oil-wet state of the core chip, 2) S at 1000 ppm changes the wettability from oil wet to water wet, 3) S at 250 ppm is intermediate, 4) many of the various ratios of S to FSF significantly change the characteristics of the core chip from oil wet to water

wet. For example, a blend of 850 ppm S and 150 ppm FSF renders the core surface water wet. Synergistic mixtures were observed between S and FSF.

**Functionalized colloidal silica (FCS).** Figures 3 and 4 show the contact angle results for various ratios of S and FCS in brine.

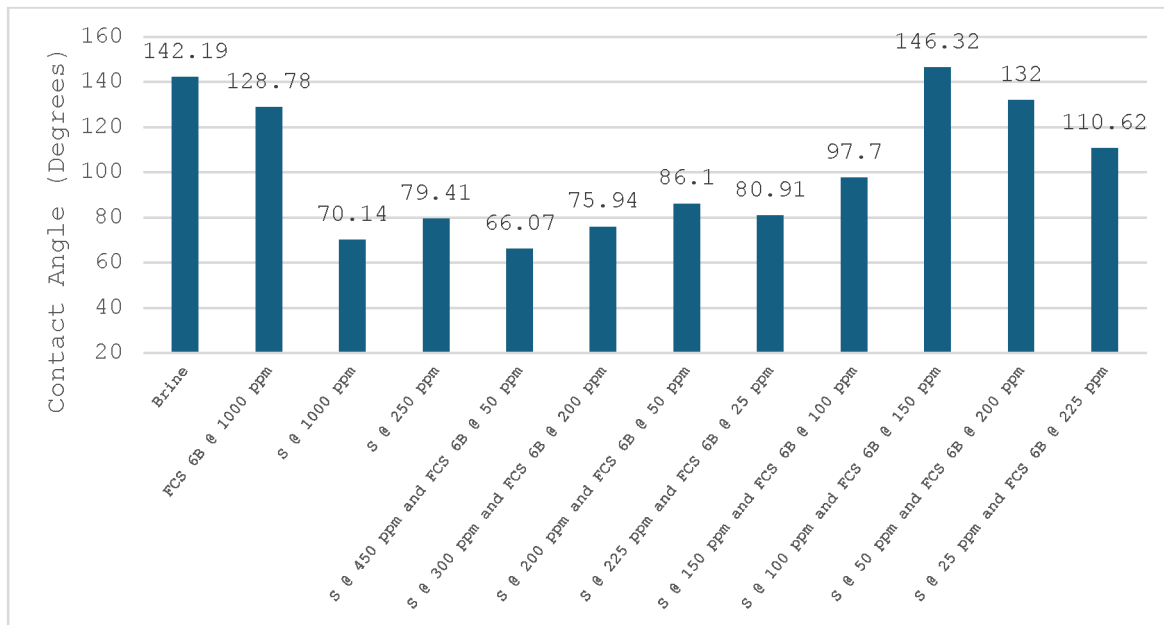


Figure 3—Contact Angle Results for Various Doses and Ratios of S and FCS 6B

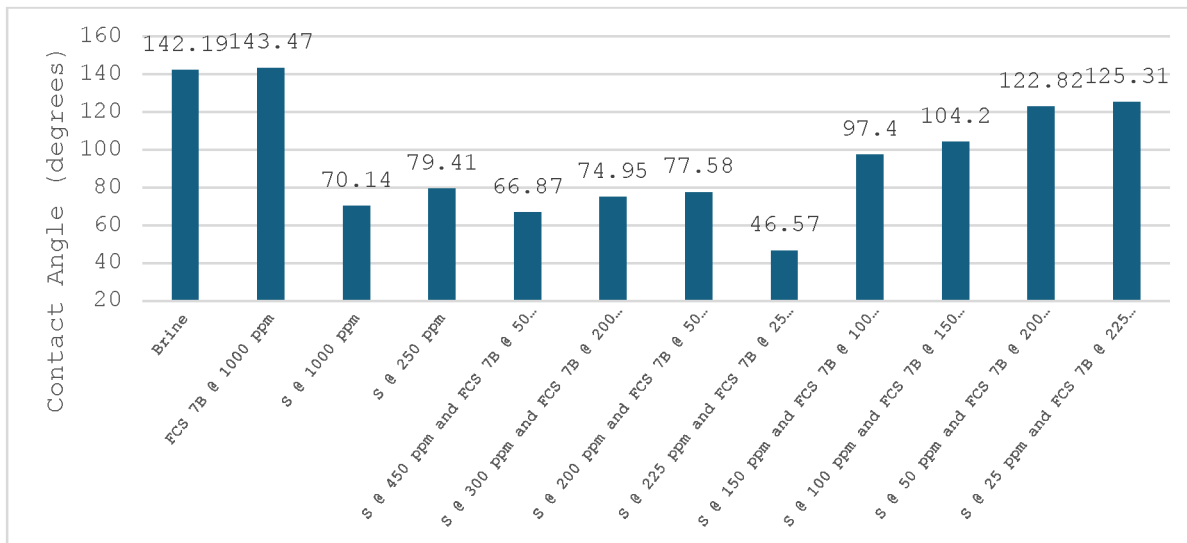


Figure 4—Contact Angle Results for Various Doses and Ratios of S and FCS 7B

As can be seen, nanoparticles alone do not lower contact angle and when compared to the brine, surfactant and blends of particles/surfactant change the wettability of the core chip from oil wet to either intermediate wet or water wet. In some cases where the concentration of the nanoparticles is higher than that of S, rock surface remains oil wet. This can be due to the adsorption of surfactant onto particles as opposed to rock surface. A further reduction in contact angle is observed for the blends of particles and surfactants compared to the effects of particles or surfactant alone. This suggests a synergistic effect arising from improved surfactant adsorption onto the rock surface which can be attributed to electrostatic repulsion by particles. In



comparison of surfactant alone at 250 ppm, a reduction of 33° and 34° is observed on the addition of 25 ppm FCS7 and 25 ppm FSF, respectively. There appears to be synergy between FCS and S. Tang et al. (2023) also reported that surfactants can also modify the surface energy and wettability of nanoparticles, enhancing their hydrophobicity to adsorb onto rock and rendering the wettability alteration. FSF, FCS6, and FCS7 were ineffective at wettability alteration and these results agreed with the literature which indicated that particles grafted with nonionic silanes are less effective than ionic ones (Alzobaidi et al., 2021). Surfactants are known to alter rock wettability. Surfactant S can form ion pairs with natural ionic surfactants in crude oil via electrostatic attraction. This will facilitate the desorption of oil wet components on rock surface, leading to rock wettability from oil-wet to water-wet or intermediate-wet as observed.

Wasan and Nikolov (2003) reported the presence of nanoparticles forming ordered layered structures in the solid-oil-aqueous phase contact region enhances the spreading of nanofluid and increases the disjoining pressure. As a result, the oil is detached from the solid surface, enhancing oil production. Hendraningrat and Torsaeter (2015) reported that wettability alteration caused by the nanoparticles, not reduction in IFT, is the dominant mechanism in improving oil recovery. Other authors concluded that, using modified SiO<sub>2</sub> and TiO<sub>2</sub>, both wettability and reduction in IFT accounted for additional oil recovery in sandstone (Ehtesabi et al., 2013; Roustaci et al., 2013).

## Interfacial tension reduction

**Functionalized fumed silica (FSF).** The results for the interfacial tension test on FSF can be seen in Figure 5. The interfacial tension between crude oil and brine was 22.25 mN/m. Addition of FSF or S reduced the IFT by 12.03 mN/m and 21.67 mN/m, respectively.

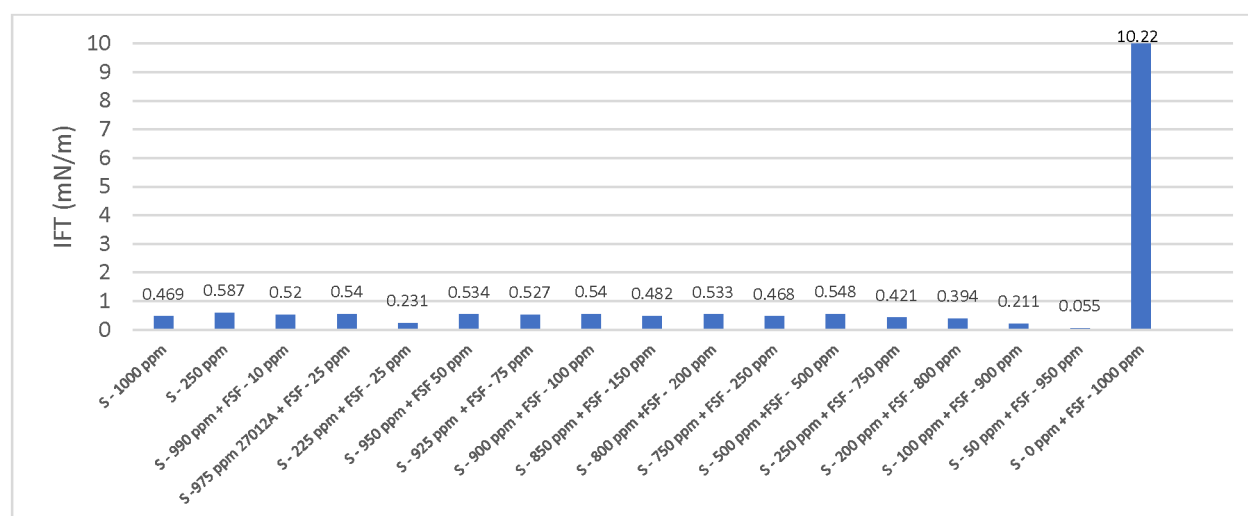
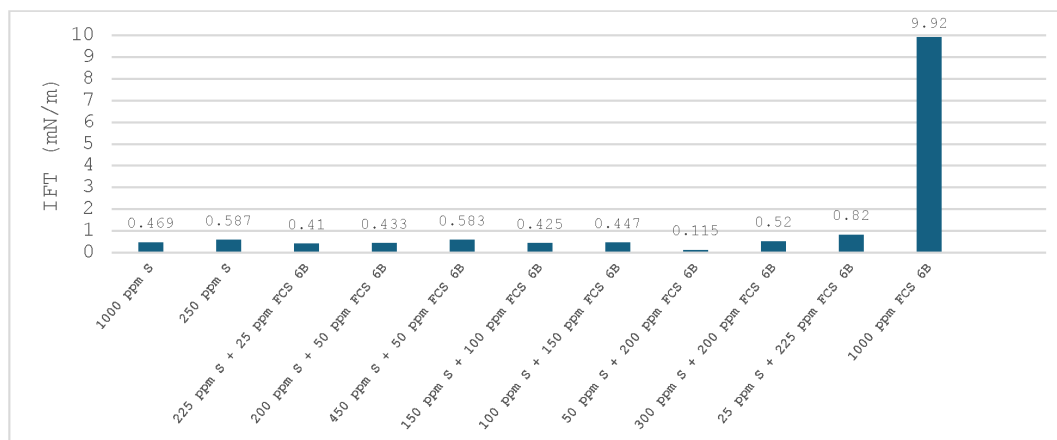


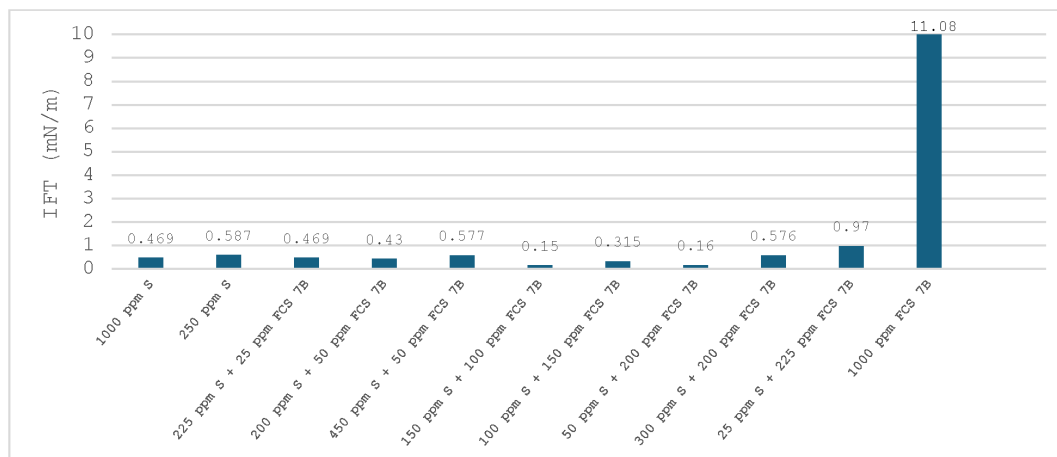
Figure 5—Interfacial Tension Results of various ratios of S to FSF in brine against crude oil. The IFT between brine and crude oil was 22.25 mN/m

The results show that all the ratios tested are either essentially equal to that of S at 1000 ppm or in some cases significantly lower. For example, a blend of 50ppm S and 950 ppm FSF exhibits an IFT of 0.055 mN/m compared to 0.469 mN/m for S alone and 10.22 mN/m for FSF alone. All cases are significantly lower than brine or FSF alone. Addition of nanoparticles screened the electrical repulsion between S molecules, allowing more S adsorbed at the oil/water interface and lowering the IFT. These results, coupled with the wettability alteration results, indicate that the synergistic mixtures of S and FSF could show an increase in oil recovery over FSF or S alone.

**Functionalized colloidal silica (FCS).** Figures 6 and 7 show the IFT results of S/FCS6 and S/FCS7 blends in brine against crude oil, respectively. Synergistic blends were observed when either 200 ppm FCS 6B or 200 ppm FCS 7B was mixed with 50 ppm S. The IFT of these blends was about 0.12 mN/m compared to 0.59 mN/m for S alone and 9.92–11.08 mN/m for nanoparticles alone. This can be due to adsorption of S onto nanoparticles, making them more hydrophobic and reducing the IFT. The synergy could be beneficial for very tight formations where low to ultralow IFT is desired.



**Figure 6—Interfacial tensions of various ratios of S to FCS 6B in brine against crude oil. The IFT between brine and crude oil was 22.25 mN/m**



**Figure 7—Interfacial tensions of various ratios of S to FCS 7B in brine against crude oil. The IFT of brine only was 22.25 mN/m**

Increasing the capillary number by lowering the interfacial tension of oil-water systems can increase oil recovery. Roustaei et al. (2012) reported the presence of polysilicon reduced the IFT from 26.3 dyne/cm to 1.75 dyne/cm, leading to additional oil recovery. It has been also shown that ZnO nanoparticles lowered the IFT between an anionic surfactant solution and oil phase drastically (Moghadam et al. 2014). They attributed this effect to adsorption of anionic surfactant on the ZnO nanoparticles, rendering them more hydrophobic and lowering the IFT faster.

### Emulsion stability

Emulsion tendency was tested at 80°C on Permian oil using synthetic brine as the dilution water. Emulsions were resolved after 20 minutes when treated with 225 ppm S and 25 ppm nanoparticles (FSF or FCS7B) compared to stable emulsion for the untreated. Addition of nanoparticles can affect the phase behavior of oil-water systems. Nanoparticles were found to stabilize emulsions which can function as efficient injecting



fluids in the reservoir and thus recovering residual oil (Kim et al., 2016); Griffith et al. 2016). However, unresolved emulsions at the wellhead will not meet transportation and production requirements of the crude oil.

## Spontaneous Imbibition

**Halloysite.** Spontaneous imbibition tests were conducted to assess the effectiveness of products on enhancing oil recovery. Surfactant adsorption onto reservoir rock surfaces can lower the economic success of enhanced oil recovery applications. Recently, a method was developed to minimize loss of surfactants by loading surfactants into the lumen of the halloysite nanotubes then coating with a thin layer of paraffin wax (Ojo, Farinmade, John, and Nguyen, 2020). The release of surfactants was controlled until this thin wax coating was in contact with oil which promotes dissolution of the wax. Fig. 8 illustrates a schematic of the preparation of wax-coated surfactant-loaded halloysites. At 70°C, tests show a 40% oil recovery for the wax-coated surfactant-loaded halloysites, compared to 16% for surfactant alone and 3% for wax-coated halloysite in the absence of surfactants (Fig. 9).

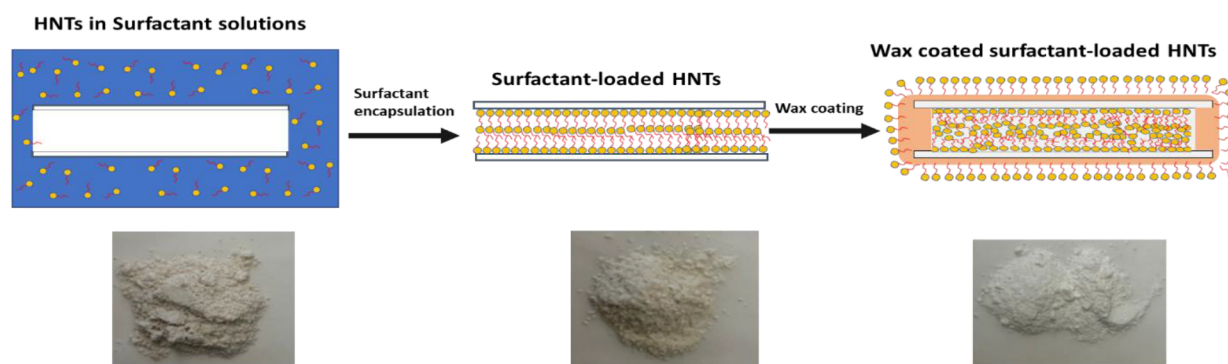


Figure 8—Schematic of the preparation of wax-coated surfactant-loaded halloysites (Ojo et al., 2020)

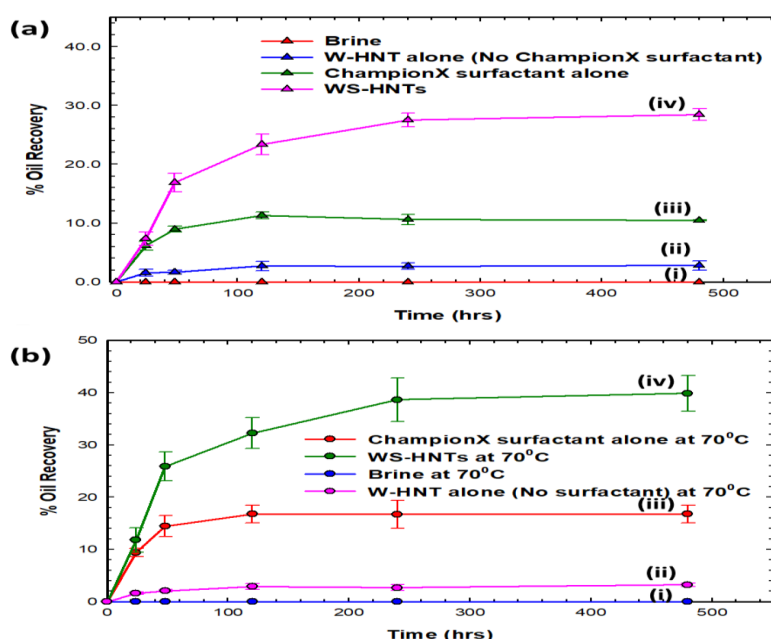


Figure 9—(a) Oil recovery from the packed capillary at ambient temperature and (b) at 70°C. (Ojo et al., 2020)

**Functionalized fumed silica (FSF).** In this study, the blends exhibiting low IFT and low contact angles ( $< 50^\circ$ ) were selected for spontaneous imbibition tests. Figure 10 shows the oil recovery by blends of S and

FSF. The results show that adding FSF to S results in a higher amount of oil being recovered than the S or FSF alone. For example, a blend of 450 ppm ZS and 50 ppm FSF (9:1 ratio) exhibited 66.7% oil recovery compared to 22.6% S at 1000 ppm alone and 10.6% FSF at 1000 ppm alone. Using 50% less chemical dose than a traditional surfactant application makes economics advantageous.

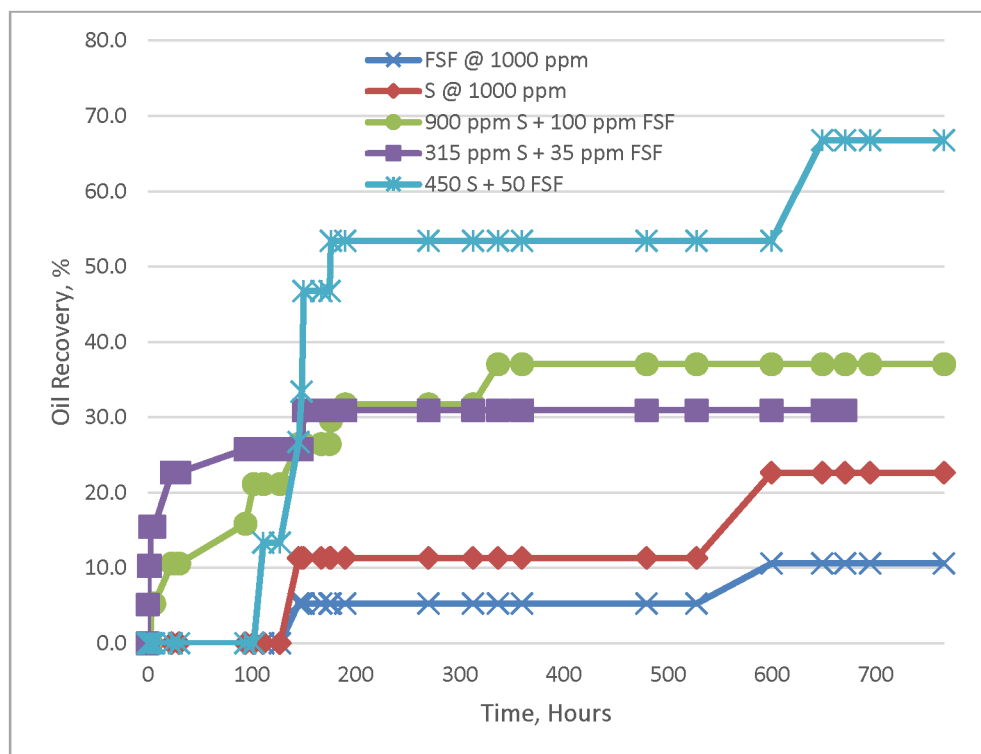


Figure 10—Spontaneous Imbibition Results for Various Doses and Ratios of S and functionalized nanoparticles

**Functionalized colloidal silica (FCF).** Figure 11 shows the spontaneous imbibition results for blends of 250 ppm S/FCS and S/FSF at 9 to 1 ratio. Even at a much lower concentration (250 ppm), the blends showed 48% more oil compared to 1000 ppm of S alone. The performances of FSF and FCS particles were comparable when blended with S.

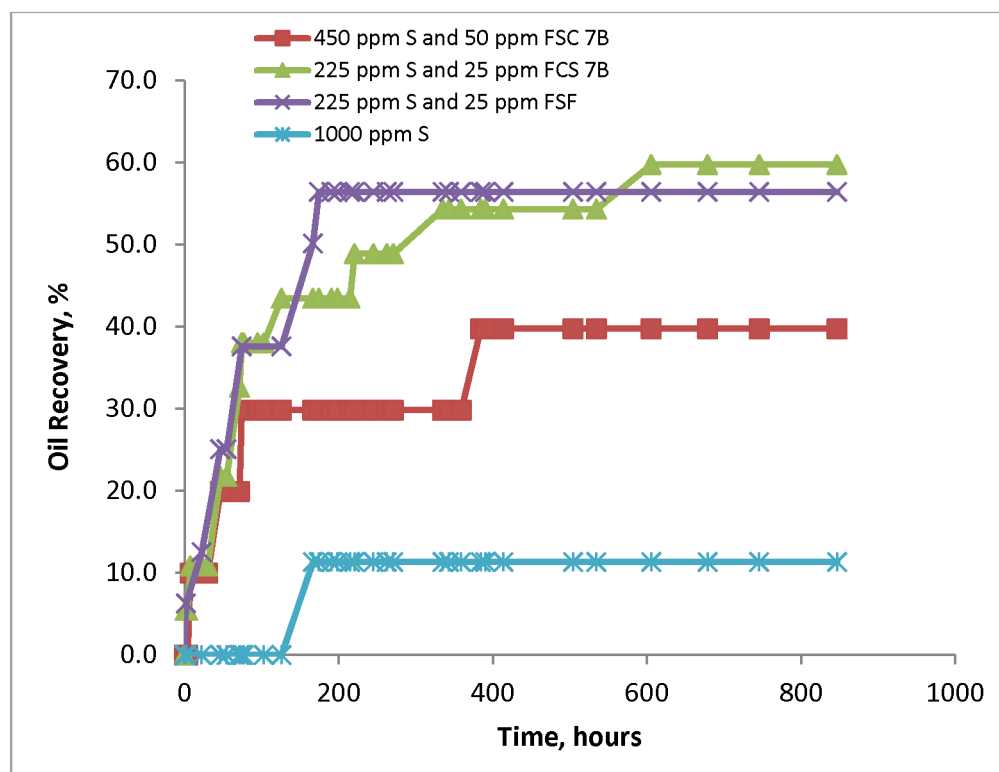


Figure 11—Spontaneous Imbibition Results for Various Doses and Ratios of S and functionalized nanoparticles

## Summary

This study presented a nanofluid composed of surface-modified colloidal nano silicas and a surfactant. Colloidal stability in brine, emulsion stability, IFT, contact angle, and spontaneous imbibition performances were evaluated to validate for the reservoir conditions. Colloidal stability was achieved and then enhanced when blended with a small amount of alum. The nanofluid effectively lowered the IFT and altered the wettability of rock surface from strongly oil wet to water-wet at a low concentration. Furthermore, the nanofluid also resolved the emulsion after 20 minutes and retrieved 48% more oil and 56% more oil compared to surfactant alone and nanoparticles alone, respectively. Therefore, it is believed that the nanofluid can play a significant role as an injection fluid to enhance oil recovery.

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